06C THERMOCHEMISTRY

UNIT 03: WHAT REACTIONS GIVE

CHEMISTRY 1310

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LEARNING OBJECTIVES

CHAPTER 6

1. Define and apply energy, work, and heat.
2. Describe and apply the concept of conservation of energy.
3. Define and apply state functions and path functions.
4. Apply the concept of state functions to predict the energy changes involved in reactions.
5. Summarize the roles of enthalpy and heat flow in chemical reactions.
6. Calculate pressure-volume work.
7. Calculate the heat required to change the temperature of a substance.
8. Calculate changes in enthalpy and internal energy in chemical reactions.
9. Apply Hess's law to calculate the enthalpy of different reactions.
10. Apply standard enthalpies of formation and Hess's law to calculate the enthalpy of a reaction.

Energy and Enthalpy

Chemistry in Context

Recall that heat (q) is the transfer of energy *due to a temperature difference* between two objects. Heat will always flow until the two objects reach the same temperature, or a state called thermal equilibrium.

You should recognize that heat is an incredibly easy quantity for us to measure in the lab because it is related to changes in temperature. All we need is a thermometer!

The next several sections are all related to *different ways* to measure or calculate ΔH_{rxn}. There are three general methods:

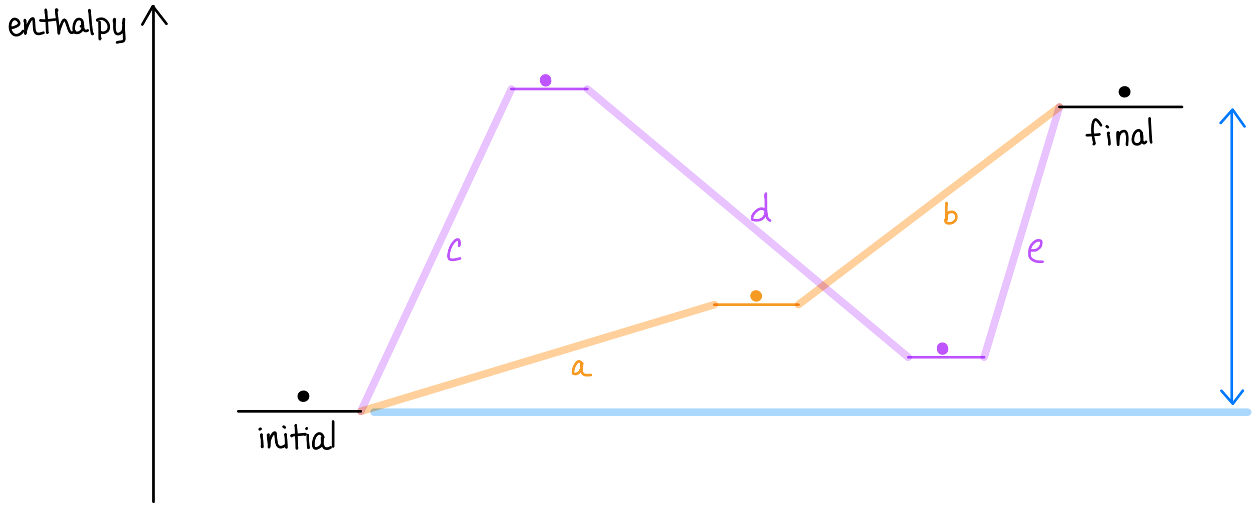
1) Experimentally through calorimetry (measuring temperature changes)
2) Hess's Law: taking advantage of state function properties
3) Theoretically through standard enthalpies of formations, $\Delta H_{\mathrm{f}}^{\mathrm{o}}$

Enthalpy in Chemical Reactions

6.7

Hess's Law

- The enthalpy change for any process is *independent* of the pathway in which the process is carried out. This works because enthalpy is a state function.



ΔH_{rxn} is the same regardless of pathway

$$\Delta H_{rxn} = H_{final} - H_{initial}$$

$$\Delta H_{rxn} = \Delta H_a + \Delta H_b$$

$$\Delta H_{rxn} = \Delta H_c + \Delta H_d + \Delta H_e$$

$$\vdots$$

- When two or more processes combine to give a resulting process, their enthalpy changes add to give the enthalpy change for the resulting process.

Enthalpy in Chemical Reactions

6.7

Hess's Law

- The overall reaction enthalpy is the *sum* of the reaction enthalpies of the steps into which the overall reaction can be divided.

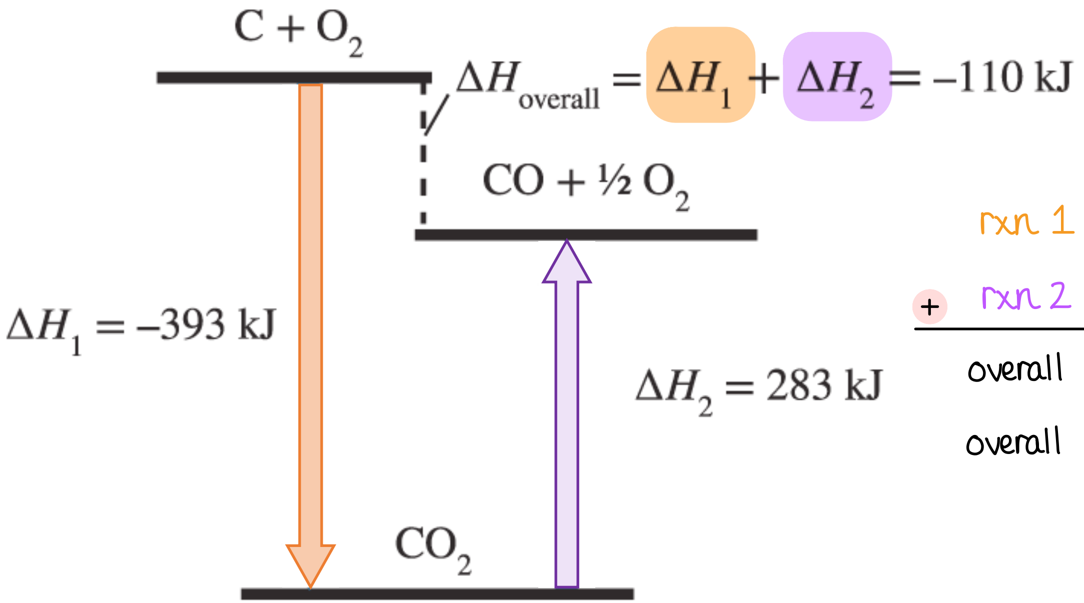


FIGURE 6.16 Diagram representing the application of Hess's Law for the overall reaction of C to CO.

rxn 1	$C + O_2 \longrightarrow CO_2$	$\Delta H_1 = -393\ kJ$
+ rxn 2	$CO_2 \longrightarrow CO + \frac{1}{2}O_2$	+ $\Delta H_2 = +283\ kJ$
overall	$C + O_2 + CO_2 \longrightarrow CO_2 + CO + \frac{1}{2}O_2$	$\Delta H_{rxn} = \Delta H_1 + \Delta H_2$
overall	$C + \frac{1}{2}O_2 \longrightarrow CO$	$\Delta H_{rxn} = -110\ kJ$

add/cancel algebraically!

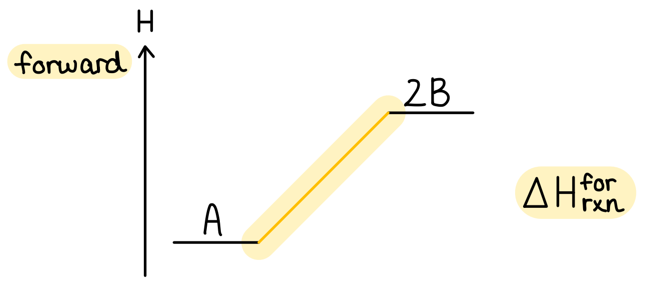
Enthalpy in Chemical Reactions

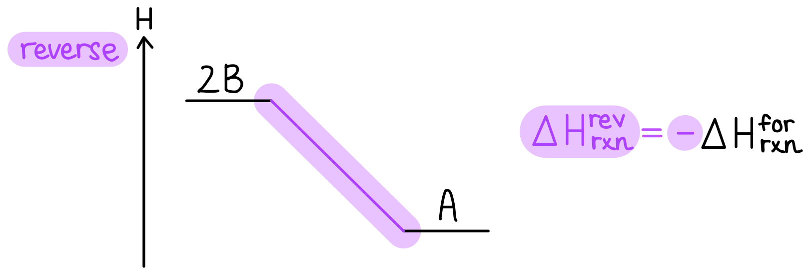
6.7

<u>General guidelines for applying Hess's Law:</u>

- Whatever you do to the reaction, do the same to its value of ΔH.
- If you reverse a reaction, change the sign of ΔH.
- If you multiply the coefficients in a reaction by a factor, multiply ΔH by the same factor.
- When you add reactions, add their ΔH.

rxn	$A \rightarrow 2B$	$\Delta H_{rxn} = +4$ kJ
reverse rxn	$2B \rightarrow A$	$-\Delta H_{rxn} = -4$ kJ
double rxn	$2A \rightarrow 4B$	$2\times \Delta H_{rxn} = +8$ kJ
quarter rxn	$\frac{1}{4}A \rightarrow \frac{1}{2}B$	$\frac{1}{4}\times \Delta H_{rxn} = +2$ kJ
doing both? triple + reverse rxn	$6B \rightarrow 3A$	$-3\times \Delta H_{rxn} = -12$ kJ





IN-CLASS QUESTION 1

Learning Objective(s): 6.9

TIP: Look for species that appear only once among rxns! e.g. BCl_3, B_2H_6, Cl_2

TIP: Align species onto "correct side" of arrow like Rubik's cube!

Use reactions **A-C** to determine the standard change in enthalpy (ΔH^o_{rxn}) for the following reaction.

$B_2H_6(g) + 6\,Cl_2(g) \rightarrow 2\,BCl_3(g) + 6\,HCl(g)$ $\Delta H^o_{rxn} = -1378.5$ kJ/mol

wrong side!

reverse, multiply by 2 **(A)** $BCl_3(g) + 3\,H_2O(\ell) \rightarrow H_3BO_3(g) + 3\,HCl(g)$ $\Delta H_A = -112.5$ kJ/mol

multiply by 12 **(B)** $\frac{1}{2}\,H_2(g) + \frac{1}{2}\,Cl_2(g) \rightarrow HCl(g)$ Don't start with HCl! $\Delta H_B = -92.5$ kJ/mol

(C) $B_2H_6(g) + 6\,H_2O(\ell) \rightarrow 2\,H_3BO_3(g) + 6\,H_2(g)$ $\Delta H_C = -493.5$ kJ/mol

$-2A$ $2\,H_3BO_3(g) + 6\,HCl(g) \longrightarrow 2\,BCl_3(g) + 6\,H_2O(\ell)$ $-2\times\Delta H^o_A = -2(-112.5\text{ kJ})$

$12B$ $6\,H_2(g) + 6\,Cl_2(g) \longrightarrow 12\,HCl(g)$ $12\times\Delta H^o_B = 12(-92.5\text{ kJ})$

$+$ C $B_2H_6(g) + 6\,H_2O(\ell) \longrightarrow 2\,H_3BO_3(g) + 6\,H_2(g)$ $+$ $\Delta H^o_C = -493.5\text{ kJ}$

$-2A+12B+C$ $B_2H_6(g) + 6\,Cl_2(g) \longrightarrow 2\,BCl_3(g) + 6\,HCl(g)$ $\Delta H_{rxn} = -2\times\Delta H^o_A + 12\,\Delta H^o_B + \Delta H^o_C$

$\Delta H_{rxn} = -1378.5\text{ kJ}$

TIP: Always add rxns to confirm!

EXAMPLE PROBLEM

Learning Objective(s): 6.9

TIP: Look for species that appear only once among rxns! Don't start with O_2!

TIP: Align species onto "correct side" of arrow like Rubik's cube!

Determine the standard change in enthalpy (ΔH^o_{rxn}) for the following reaction.

$4\ FeO(s) + O_2(g) \rightarrow 2\ Fe_2O_3(s)$ $\qquad \Delta H^o_{rxn} = -560\ kJ/mol$

(A) $2\ Fe(s) + O_2(g) \rightarrow 2\ FeO(s)$ wrong coefficient! wrong side! $\qquad \Delta H^o_A = -544\ kJ/mol$ reverse, multiply by 2

(B) $4\ Fe(s) + 3\ O_2(g) \rightarrow 2\ Fe_2O_3(s)$ $\qquad \Delta H_B = -1648\ kJ/mol$

$-2\times A$: $4FeO(s) \rightarrow 4\ Fe(s) + 2O_2(g)$ $\qquad -2\times \Delta H_A = -2\times(-544\ kJ)$

$+$ B : $4Fe(s) + 3O_2(g) \rightarrow 2\ Fe_2O_3(s)$ $\qquad + \ \Delta H_B = -1648\ kJ$

$-2\times A + B$: $\cancel{4Fe(s)} + 4FeO(s) + 3O_2(g) \rightarrow \cancel{4Fe(s)} + 2\ Fe_2O_3(s) + 2O_2(g)$

cancels

simplifies

$4FeO(s) + O_2(g) \rightarrow 2\ Fe_2O_3(s)$

$\Delta H_{rxn} = -2\times \Delta H_A + \Delta H_B$

$= -2\times(-544\ kJ) + (-1648\ kJ)$

$\Delta H_{rxn} = -560\ kJ$

TIP: Always add rxns to confirm!

Standard Enthalpies of Formation

6.8

Standard enthalpies/heats of formation (ΔH_f^o)

a) *What is a standard formation reaction?*

- The formation of 1 mole of a substance (in a specified state) from its constituent elements in their standard states.
- Because the definition specifies only 1 mole of product, fractional coefficients are allowed for the reactants (the constituent elements).

? elements ⟶ 1 compound (state)

b) *What are standard states of elements?*

- The *most energetically stable* form of the elements under standard conditions.
- Standard conditions are denoted by the ° symbol and *defined* as 1 atm for gases and 1 M for aqueous solutions. It also often *implies* 25 °C.
  - Metals → most written as monatomic solids $Fe(s)$, $Na(s)$, etc.
  - Nonmetals → recall the diatomics $H_2(g)$, $O_2(g)$, $N_2(g)$, $Cl_2(g)$, $F_2(g)$, $Br_2(l)$, $I_2(s)$

    → monatomic solids $C(s)$, $Si(s)$

$\Delta H_f^o = \emptyset$ by definition; i.e. "free"

Standard Enthalpies of Formation

6.8

Standard enthalpies/heats of formation (ΔH_f^o)

- The enthalpy change associated with the formation of 1 mole of a substance (in a specified phase) from its constituent elements in their standard states.

$\Delta H_f^o = \emptyset$ ΔH_f^o

? elements $\rightarrow$ 1 compound (state)

$$C(s) + O_2(g) \rightarrow 1\,CO_2(g) \qquad \Delta H^o_{f,CO_2(g)}$$

$$Zn(s) + Cl_2(g) \rightarrow 1\,ZnCl_2(s) \qquad \Delta H^o_{f,ZnCl_2(s)}$$

$$H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow 1\,H_2O(l) \qquad \Delta H^o_{f,H_2O(l)}$$

$$Pb(s) + N_2(g) + 3O_2(g) \rightarrow 1\,Pb(NO_3)_2(s) \qquad \Delta H^o_{f,Pb(NO_3)_2(s)}$$

- Definition implies any element in its standard state must have $\Delta H_f^o = 0$.
- A table of standard heats of formation for various compounds can be found in Table 6.4 and Appendix A.2 of your textbook.

  These values will be provided for you when needed.

Standard Enthalpies of Formation

6.8

Applying Hess's Law, the standard enthalpies of formation (ΔH_f^o) can be used to calculate the standard change in enthalpy for any reaction (ΔH_{rxn}^o).

- This is the *easiest* method to calculate ΔH_{rxn}^o!

$$aA + bB \longrightarrow cC + dD$$

$$\Delta H^o_{rxn} = ?$$

coefficients from balanced chemical equation

$$\Delta H^o_{rxn} = \sum n_P \cdot \Delta H^o_{f,P} - \sum n_R \cdot \Delta H^o_{f,R}$$

products reactants

$$\Delta H^o_{rxn} = (c \cdot \Delta H^o_{f,C} + d \cdot \Delta H^o_{f,D}) - (a \cdot \Delta H^o_{f,A} + b \cdot \Delta H^o_{f,B})$$

EXAMPLE PROBLEM

Learning Objective(s): 6.10

Calculate ΔH^o_{rxn} for the following reaction using the ΔH^o_f data below.

$C_3H_8(g) + 5\ O_2(g) \rightarrow 3\ CO_2(g) + 4\ H_2O(\ell)$

$\Delta H^o_{rxn} =$ ____________

element in standard state!

$\Delta H^o_f = 0$

$$\Delta H^o_{rxn} = \sum n_P \cdot \Delta H^o_{f,P} - \sum n_R \cdot \Delta H^o_{f,R}$$

$$= \left[3 \cdot \Delta H^o_{f,CO_2} + 4 \cdot \Delta H^o_{f,H_2O}\right] - \left[1 \cdot \Delta H^o_{f,C_3H_8} + 5 \cdot \Delta H^o_{f,O_2}\right]$$

$$= \left[3 \cdot \left(-395.3\ \frac{kJ}{mol}\right) + 4 \cdot \left(-285.8\ \frac{kJ}{mol}\right)\right] - \left[1 \cdot \left(-103.8\ \frac{kJ}{mol}\right) + 5 \cdot \left(0\ \frac{kJ}{mol}\right)\right]$$

$$\Delta H^o_{rxn} = -2225.3\ \frac{kJ}{mol}$$

Substance	$\Delta H^o_f \left(\frac{kJ}{mol}\right)$
$C_3H_8(g)$	−103.8
$CO_2(g)$	−395.3
$H_2O(\ell)$	−285.8

IN-CLASS QUESTION 2

Learning Objective(s): 6.10

Calculate ΔH^{o}_{rxn} for the following reaction using the ΔH^{o}_{f} data below.

$$P_4O_{10}(s) + 6\ H_2O(\ell) \rightarrow 4\ H_3PO_4(aq) \qquad \Delta H^{o}_{rxn} = \underline{\qquad\qquad}$$

$$\Delta H^{o}_{rxn} = \sum n_P \cdot \Delta H^{o}_{f,P} - \sum n_R \cdot \Delta H^{o}_{f,R}$$

$$= \left[4 \cdot \Delta H^{o}_{f,H_3PO_4}\right] - \left[1 \cdot \Delta H^{o}_{f,P_4O_{10}} + 6 \cdot \Delta H^{o}_{f,H_2O}\right]$$

$$= \left[4 \cdot \left(-1288.34\ \frac{kJ}{mol}\right)\right] - \left[1 \cdot \left(-2984\ \frac{kJ}{mol}\right) + 6 \cdot \left(-285.8\ \frac{kJ}{mol}\right)\right]$$

$$\Delta H^{o}_{rxn} = -454\ \frac{kJ}{mol}$$

Substance	$\Delta H^{o}_{f}\ \left(\frac{kJ}{mol}\right)$
$P_4O_{10}(s)$	−2984
$H_2O(\ell)$	−285.8
$H_3PO_4(aq)$	−1288.34

BONUS QUESTION

Assessed on: *undetermined*

Use reactions **A-D** to determine the standard change in enthalpy (ΔH^{o}_{rxn}) for the following reaction.

$C_6H_4(OH)_2(aq) + H_2O_2(aq) \rightarrow C_6H_4O_2(aq) + 2\ H_2O(\ell)$ $\quad \Delta H^{o}_{rxn} =$ ____________

(A) $C_6H_4(OH)_2(aq) \rightarrow C_6H_4O_2(aq) + H_2(g)$ $\quad \Delta H_{\mathbf{A}} = +177.4\ \mathrm{kJ/mol}$

(B) $H_2(g) + O_2(g) \rightarrow H_2O_2(aq)$ $\quad \Delta H_{\mathbf{B}} = -191.2\ \mathrm{kJ/mol}$

(C) $H_2(g) + \frac{1}{2}\ O_2(g) \rightarrow H_2O(g)$ $\quad \Delta H_{\mathbf{C}} = -241.8\ \mathrm{kJ/mol}$

(D) $H_2O(g) \rightarrow H_2O(\ell)$ $\quad \Delta H_{\mathbf{D}} = -43.8\ \mathrm{kJ/mol}$